Aditya Shibu

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Dubai, UAE

Summary

CS/AI graduate with hands-on research experience across the full robotics stack: perception, planning, control, and edge deployment. Built and optimized real-time inference pipelines in C++, CUDA, and TensorRT on NVIDIA Jetson and L40s hardware. Two published conference papers and a journal preprint targeting RA-L. 1st Place at Formula Student AI.

Education

BSc (Hons) Computer Science with Artificial Intelligence — **GPA: 4.0/4.0, First Class Honours**
Heriot Watt University, Dubai, UAE — *Sept 2023 - Jul 2026*

Research and Publications

- **Simulating an Autonomous System in CARLA Using ROS2** — **Research Paper** - Published
  International Conference on Signal Processing and Information Security (ICSPIS) — *November 2025*
- **SkySim: Natural Language Control of Drone Swarms using LLMs** — **Research Paper** - Published
  International Conference on Drones and Unmanned Systems (DAUS) — *February 2026*
- **LiZIP: A Deep Learning Model for Compression of LiDAR point clouds** — **Research Paper** - Preprint
  School of Mathematical and Computer Sciences - Heriot Watt University — *Target - RA-L*

Skills

- **Languages**: C, C++, Python, CUDA, JavaScript, Java, OCaml
- **AI / Robotics**: Computer Vision, OpenCV, Open3D, NumPy, LiDAR, point clouds, sensor fusion, coordinate transformations (tf), GNSS/INS (RTK, NTRIP), Data loaders, KITTI, nuScenes, Argoverse, PyTorch, TensorFlow, TensorRT, YOLO, ROS 2, rosbag, CARLA, Gazebo, Ollama
- **Tools**: Linux, Git, GitHub Actions, Docker

Work Experience

- **Center for AI and Robotics (CAIR) - New York University** — Abu Dhabi, UAE
  Research Collaborator (Remote / Part-Time) — *Dec 2025 - Feb 2026*
  - Architected and trained a **Conditional Imitation Learning (CILRS)** neural network using PyTorch to navigate complex urban intersections based on high-level GPS commands.
  - Configured and managed headless CARLA simulations on dual **NVIDIA L40s HPC** infrastructure, building synthetic urban driving datasets (RGB + Telemetry) for end-to-end driving models.

- **CanopyAI** — Dubai, UAE
  Forward Deployed AI Engineer (Freelance) — *Oct 2026 - Present*
  - Architecting a **PPE detection and color-coding** computer vision pipeline using **YOLO26 with SAHI** (Slicing Aided Hyper Inference) on an **NVIDIA Blackwell** GPU, for integration into a software platform serving clients across retail, hospitality, mining and construction.
  - Deployed on-site edge AI infrastructure for **Dubai Holding (Nakheel)**, flashing and configuring 4 **NVIDIA Jetson AGX Thor/Orin** units and an **RTX PRO 6000** workstation, with network configuration and multi-camera integration.

- **Lab for AI and Robotics Research (LAIRR) - Heriot Watt University** — Dubai, UAE
  Research Intern (Hybrid / Part-Time) — *Nov 2025 - Present*
  - Currently leading a team on a hardware-tested drone swarm benchmark study with **Prof. Claudio Zito**, comparing formation control, shape assignment, dropout recovery, and drone coordination on **Bitcraze Crazyflie** swarms, targeting a review and experimental benchmark paper.
  - Engineered **LiZIP**, a novel auto-regressive neural compression architecture for LiDAR point clouds, with the full inference pipeline written from scratch in **C++**, custom **CUDA** kernels, and **TensorRT**, and the model developed in **PyTorch**.
  - Built **PyTorch data loaders** and preprocessing pipelines for **KITTI**, **nuScenes**, and **Argoverse**, training LiZIP on nuScenes and generalizing to Argoverse with even higher compression (payloads of **605 KB on nuScenes and 186 KB on Argoverse**), outperforming **LASzip and Google Draco** at a near-lossless maximum error of **<0.017 mm**, and benchmarking on KITTI, with **Open3D** used to inspect reconstructed **point clouds**.
  - Optimized for bare-metal edge deployment on the NVIDIA Jetson AGX Orin, achieving real-time inference latency of **16.7ms to 26.8ms** per frame, **outperforming LASzip by up to 2x in speed** while consuming <20% CPU and <1% GPU utilization.
  - Built the perception system for **CogWaters**, a multi-institutional **Dubai RDI-funded** project (Heriot-Watt Dubai, Edinburgh, UK National Robotarium) developing cognitive architectures for autonomous debris-collecting USVs: collected and annotated simulation datasets, trained a **YOLOv8** model, and optimized inference with **TensorRT** for real-time detection.

- **AimbeyonD - Heriot Watt University** Dubai, UAE
  Research Technician (On-Site / Part Time) *Nov 2025 - Present*
  - Currently leading a team to distill an LLM into a small language model that runs locally on lab machines and is spatially aware, extending natural language control from static shapes to drone swarm formations.
  - Developed **SkySim**, a ROS 2 Jazzy-based simulation environment on Gazebo for Natural Language Control of Drone Swarms using LLMs, with **Artificial Potential Fields**-based collision avoidance and formation flying
  - Deployed SkySim on **10x Bitcraze Crazyflie 2.1 Brushless** drones using OptiTrack motion capture for real-time position streaming to the SLM, enabling natural language control of real drone swarms in static formations.
  - Led development of **Autonomous Underground Infrastructure Intelligence System (AUIIS)** in collaboration with the **Roads and Transport Authority (RTA) Dubai** for the **Dubai RDI Fund**.

- **Atlas Racing FS** Dubai, UAE
  Head of Driverless & Lead Autonomous Systems Developer (On-Site / Part-Time) *May 2025 - July 2026*
  - Built a multi-sensor perception pipeline in **ROS 2** using **ZED2i stereo cameras and RoboSense Helios 16 LiDAR**, implementing **DBSCAN clustering** and camera-LiDAR sensor fusion for object detection and localization, defining the vehicle **tf frames** with **coordinate transformations** from the LiDAR and camera frames to the vehicle frame, and building an **OpenCV** HUD to display YOLO health and live detections
  - Recorded and replayed **ROS bags** to debug the perception pipeline and investigate a delayed steering response on the car from logged sensor and control data
  - Integrated a **CHCNAV GNSS/INS** into the ROS 2 stack over Ethernet (RJ45) using the **nmea_navsat_driver** package, configured **NTRIP** for **RTK** corrections, and wrote custom scripts to auto-connect and monitor **GNSS** stream health
  - Optimized **YOLO inference with TensorRT acceleration**, reducing latency by **88%** (140ms → 16ms) and achieving stable **40-50 FPS** for real-time autonomous vehicle operation
  - Led technical development resulting in **1st Place Overall Statics** and **1st Place Real World AI**, demonstrating production-ready autonomous systems performance
  - Scaled the driverless division from a **5-person founding team to 60+ members across design, outreach, and engineering subsections**, establishing organizational structure and mentoring new recruits.

Projects

- **Kernel Intelligence through Expert Fine-tuning (KITE)**: A fine tuned Small Language Model (SLM) that has been trained on a handmade dataset of Python and CUDA pairs to train the SLM to convert unoptimized Python code to optimized CUDA code | **Tech: Python, OpenAI Triton, CUDA** (May 2025) | **To be released soon**
- **Hydrakon V1 and V2**: Worked with a team of 9 to build an autonomous racing car (Simulation and Hardware Integration) for Formula Student UK | **Tech: CARLA, YOLO, ROS2, Cartographer, Python, C** (April 2025) | **https://github.com/Atlas-Racing**
- **Local RAG**: Developed a fully local RAG with Ollama and LangChain | **Tech: Python, Ollama, and LangChain** (January 2025) | **https://github.com/jonathanjthomas/GDG-RAG-Demo**
- **Space Debris Classification**: Developed a space debris classification model using YOLO, enhancing training data by combining datasets from Roboflow for broader coverage | **Tech: Python, Jupyter Notebook, YOLO** (December 2024) | **https://github.com/adityashibu/spaceDebrisDetection**

Awards and Certifications

- **A Beginner's Guide to Autonomous Robots** - NVIDIA - March, 2026
- **Babcock FS-AI Overall Statics Award Winner** - IMechE - July, 2025
- **FS-AI Real World Award Winner** - IMechE - July, 2025
- **Deep Learning Essentials Intermediate** - IBM - January, 2025
- **GitHub Foundations certification** - GitHub - December, 2024

Leadership and Volunteering Experience

- **Google Developer Groups** Dubai, UAE
  Lead of AI and ML *May 2025 - July 2026*
  - Promoted from Co-lead to Lead after organizing a successful workshop with **120+ attendees**, demonstrating exceptional event management and technical presentation skills

- **Heriot Watt University** Dubai, UAE
  MACS Department Officer *August 2025 - July 2026*
  - Lead a team of **12 class representatives** across all MACS courses, serving as primary student advocate in faculty committees and implementing policy changes impacting **600+ students**

- **Heriot Watt University** Dubai, UAE
  Lead Copy Editor and Typesetter, International Social Capital Conference *September 2026 - Present*
  - Lead copy editing and serve as the main typesetter for Volume 1 of the *Proceedings of the International Social Capital Conference* (Social Capital 2026, International Social Capital Association), building the publication format and layout used for the papers